

Task 57: Number of Provinces

Problem Description

There are n cities. Some of them are connected, while some are not. If city a is connected directly with city b , and city b is connected directly with city c , then city a is connected indirectly with city c . A province is a group of directly or indirectly connected cities and no other cities outside of the group. You are given an $n \times n$ matrix `isConnected` where `isConnected[i][j] = 1` if the i th city and the j th city are directly connected, and `isConnected[i][j] = 0` otherwise. Return the total number of provinces.

Java Solution

```
class Solution {
    public int findCircleNum(int[][] isConnected) {
        int n = isConnected.length;
        boolean[] visited = new boolean[n];
        int provinces = 0;

        for (int i = 0; i < n; i++) {
            if (!visited[i]) {
                provinces++;
                dfs(isConnected, visited, i);
            }
        }

        return provinces;
    }

    private void dfs(int[][] isConnected, boolean[] visited, int i) {
        visited[i] = true;
        for (int j = 0; j < isConnected.length; j++) {
            if (isConnected[i][j] == 1 && !visited[j]) {
                dfs(isConnected, visited, j);
            }
        }
    }
}
```